

Evaluation of validation strategies and transferability of EEG attention decoding: a comparative analysis on open datasets

Sanzhar Sovet¹, Beibit Abdikenov^{1*}, Manzura Zholdasova^{2,3}, Altyngul Kamzanova², Medet Mukushev¹, Almira Kustubayeva^{2,3}

1 Science and Innovation Center “Artificial Intelligence”, Astana IT University, Astana, Kazakhstan

2 Department of Biophysics, Biomedicine, and Neuroscience, Al-Farabi Kazakh National University, Almaty, Kazakhstan

3 Brain Institute, Al-Farabi Kazakh National University, Almaty, Kazakhstan

* beibit.abdikenov@astanait.edu.kz

Abstract

[TODO — Write the abstract last. Limit: 300 words.]

Introduction

Electroencephalography (EEG) is a technique for recording brain activity using multichannel scalp electrodes that measure voltage signals reflecting underlying cognitive processes. It is widely used in neuroscience and brain-computer interface (BCI) research due to its non-invasive nature, high temporal resolution, portability and relatively low cost [1]. At the same time, EEG signals are inherently noisy, non-stationary, and exhibit high inter-subject and inter-session variability [2]. In addition, these challenges are compounded by the scarcity of large, diverse, and well-annotated open EEG datasets [3]. Consequently, developing a robust decoding model with strong generalization capabilities across subjects, sessions and tasks remains an open and central challenge [2, 4]. One promising approach to solve these problems is transfer learning (TL). TL reuses representations already learned on one or more source problems to reduce the amount of labeled data and training effort a related target problem would otherwise require [5]. Its success depends not only on modeling and learning strategies but also on the degree of domain and label compatibility between source and target datasets.

For cognitive constructs such as attention, this compatibility is especially consequential and difficult to guarantee because attention has always been an ambiguous concept and is not a unitary construct with a single universally accepted operational definition [3, 6]. Rather, attention is an umbrella term encompassing a set of partially independent processes and functions, such as selecting relevant information, maintaining focus over time, shifting attention between tasks, and allocating cognitive resources. These processes may operate on both external sensory input and internally represented information, including information maintained in working memory [6–8]. The relative contributions of these processes may vary with task demands, sensory modalities, task goals, experimental instructions, assigned conditions, self-reported assessments, and other aspects of experimental design and label definition. Consequently, experimental paradigms described as measuring “attention” may probe non-equivalent configurations of cognitive processes. As a result, different datasets may represent different attention-related cognitive states while assigning them the same nominal labels.

Previous EEG transfer-learning research has been predominantly concerned with cross-subject and cross-session generalization, whereas cross-task, cross-dataset, and cross-paradigm transfer have received comparatively less systematic attention [2, 9, 10]. Consequently, systematic empirical evidence for source–target suitability across heterogeneous attention-related EEG datasets and tasks remains limited. Source–target suitability is often motivated conceptually or inferred from nominally aligned labels. This leaves unclear which source–target combinations provide useful zero-shot transfer, how much target

discriminability is retained relative to target-specific baselines, and whether transferability is direction-dependent. Establishing such pre-adaptation baselines is necessary to justify source selection and to interpret gains or negative transfer from subsequent transfer-learning and domain-adaptation methods [11].

[IN PROGRESS — This paragraph is retained provisionally. Remove it if suitable experiments comparing operational label mappings cannot be completed.]

Here, we contribute to addressing this gap by systematically evaluating zero-shot generalization across two technically compatible public EEG datasets spanning multiple tasks and paradigms. We examine both within-dataset baseline performance and transfer across subjects, sessions, trials, tasks, and datasets under a common preprocessing pipeline, compare directed source–target pairs against target-specific baselines, and assess transferability using five decoders ranging from linear and tree-based baselines to deep neural networks. Together, these analyses yield reproducible pre-adaptation benchmarks for evaluating candidate sources and contextualizing both gains and negative transfer produced by subsequent transfer-learning and domain-adaptation methods.

Open-access datasets

EEG data for mental attention state detection (Dataset A)

The dataset described in [12] comprises **34 EEG recordings** collected from **five participants** over the course of **seven days**, with the fifth participant completing only **six sessions**. During each session, participants operated a virtual train using the *Microsoft Train Simulator*. Each recording lasted approximately **35 to 55 minutes**. All experiments in this study were performed with healthy volunteer participants chosen among students.

To ensure consistency, all sessions were conducted on the same predefined route, designed to require minimal interaction. This setup constitutes a **low-intensity control task**, minimizing variability due to task demands and aligning with *naturalistic stimulus-based protocols* as described in [13]. Each session was divided into **three sequential phases**, each designed to reflect a different target mental state: ***focused***, ***unfocused***, and ***drowsy***. The participants simulated these mental states based on standardized instructions provided by the experiment supervisor:

- **Focused (0–10 min):** Participants actively monitored and mentally engaged with the simulator, maintaining sustained attention to the virtual controls and screen events.
- **Unfocused (10–20 min):** Participants disengaged from the task, ceasing interaction and intentionally

avoiding attentional focus, while being instructed to keep their eyes open and remain awake.

- **Drowsy (20–30 min):** Participants were allowed to relax, close their eyes, and enter a drowsy or lightly dozing state, simulating reduced cognitive arousal.

EEG signals were recorded using a **modified Emotiv headset**, equipped with wet electrodes and supporting wireless Bluetooth transmission. The device recorded data at a **sampling rate of 128 Hz** and with a **voltage resolution of 0.51 μV** . The dataset includes **14 EEG channels**, labeled according to the 10–20 international system: AF3, F7, F3, FC5, T7, P7, O1, O2, P8, T8, FC6, F4, F8, and AF4.

SAM-40: dataset of 40-subject EEG recordings (Dataset B)

The **SAM-40** dataset, described in [14], contains EEG recordings from 40 healthy participants (14 females and 26 males) with a mean age of 21.5 years. The purpose of data collection was to investigate short-term stress responses induced by the performance of various cognitive tasks, which correspond to a *continuous cognitive engagement* paradigm as described in [13]. The experimental protocol included four conditions:

- **Resting state** — EEG recording during relaxation while listening to music.
- **Stroop Color-Word Test** — participants were required to identify the color of the ink in which a word was printed, while ignoring the word’s semantic meaning. The task included two types of stimuli: *congruent* (the color name matched the ink color) and *incongruent* (the color name and ink color differed). Each execution of the task included 11 stimuli.
- **Arithmetic problem solving** — participants mentally solved a mathematical problem and determined whether the presented answer was correct, responding with a “thumbs up” or “thumbs down” gesture. Each execution of the task included 6 arithmetic stimuli.
- **Mirror Image Recognition** — participants were shown images and asked to determine whether they were mirror-symmetric or asymmetric, responding with the same gestures. Each execution of the task included 8 images.

Each task lasted 25 seconds, with 5-second rest intervals and a 10-second instruction period preceding the next task. After the completion of each trial, participants rated the level of stress experienced for each task on a 10-point scale, where 1 indicated minimal stress and 10 indicated maximal stress. For each participant, three trials were recorded using different sets of stimuli.

EEG signals were recorded using an Emotiv Epoc Flex headset with 32 wet electrodes, labeled according to the international 10–20 system. Data were acquired at a sampling rate of 128 Hz and a voltage resolution of 0.51 μ V. The 32 channels included in the dataset are: Cz, Fz, Fp1, F7, F3, FC1, C3, FC5, FT9, T7, CP5, CP1, P3, P7, PO9, O1, Pz, Oz, O2, PO10, P8, P4, CP2, CP6, T8, FT10, FC6, C4, FC2, F4, F8, and Fp2.

Methodology

The methodological design of this study was developed to evaluate the generalizability and transferability of EEG-based attention decoding across different recording conditions, participants, and task paradigms. To this end, we adopted two publicly available datasets that differ in experimental protocols but share compatible technical specifications, which allows for a controlled comparative analysis. The datasets were harmonized through a binary operational labeling scheme that distinguishes conditions of high protocol-defined task demand from conditions of rest or task disengagement, enabling consistent evaluation across scenarios. This section outlines the rationale for dataset alignment, the strategies employed for cross-validation, and the experimental settings designed to assess within-dataset performance as well as cross-subject, cross-session, cross-task, and cross-dataset generalization.

All experiments were executed through a version-controlled computational workflow. For every run, the workflow stores the resolved configuration of data preparation, validation and model settings together with the run outputs needed to reconstruct the reported evaluation metrics. These records provide run-level provenance for the results and permit the correspondence between a reported metric and the experiment that produced it to be inspected. The complete implementation, experiment configurations and run-level artifacts are available at <https://github.com/sovetskiysn/eeg-validation-strategies>.

Dataset alignment and integration rationale

Effective evaluation of cross-task and cross-dataset generalization relies on the availability of datasets that are compatible both technically and conceptually. In this study, we integrate two publicly available EEG datasets that differ in their original purpose and labeling schemes, yet retain sufficient technical and conceptual compatibility for a meaningful comparison.

A major strength of this dataset combination lies in their acquisition setup. Both datasets were recorded using Emotiv EEG systems with the same sampling rate (128 Hz) and voltage resolution, although their original electrode montages differed (14 versus 32 channels). For cross-dataset analyses, only the 12 channels common to both datasets were retained. Differences in EEG recording systems and electrode configurations

can hinder cross-dataset transferability [15]. Restricting the analysis to the shared channels substantially reduces this configuration-related mismatch, although it does not eliminate all acquisition-related distribution shifts. Within the limited set of publicly available datasets suitable for this comparison, this pair offered a comparatively compatible basis for the present comparative evaluation.

The conceptual rationale for combining the datasets rests on the cognitive conditions represented in their respective protocols. The mental attention-state dataset contains explicit attention-state labels (*Focused*, *Unfocused*, and *Drowsy*) collected during a low-interaction simulation task, whereas SAM-40 was originally designed for stress assessment. Both capture cognitive conditions strongly tied to attentional processes. The SAM-40 dataset includes tasks widely recognized in cognitive neuroscience for eliciting high levels of attentional engagement:

- **Stroop Color Test** — probes both *selective* and *sustained attention*, requiring participants to suppress automatic reading responses and continually focus on the ink color throughout the task [16–18].
- **Mental arithmetic task** — demands *sustained internal attention* and working memory for continuous manipulation of numerical information [19, 20].
- **Mirror image recognition task** — engages visuospatial processing and requires both *selective* and *sustained attention* to discriminate between symmetric and asymmetric patterns, relying on mental rotation processes that demand continuous cognitive effort [21–23].

The simulator-based and task-based datasets correspond, respectively, to the *naturalistic stimulus-based protocols* and *continuous cognitive engagement* paradigms described in [13]. In the unified labeling scheme, *Rest* and *Unfocused* are treated as low-attention states, whereas *Focused*, *Stroop*, *Arithmetic*, and *Mirror recognition* are treated as high-attention states. The *Drowsy* phase is excluded from this mapping and from all subsequent preprocessing and analyses because permitted eye closure and drowsiness may confound the comparison with differences in arousal, visual input, and ocular activity. This mapping is summarized in Table 1.

Table 1. Unified binary attention label mapping across datasets.

Attention State	Conditions
High Attention	Focused, Stroop, Arithmetic, Mirror
Low Attention	Unfocused, Rest

Validation strategies

We evaluated the decoders under five validation protocols, each addressing a distinct generalization question. To preserve the independence of the training and test data, all windows originating from the same recording unit were assigned to the same partition. This prevents overlapping windows from being divided between training and test sets, reducing the risk of overly optimistic performance estimates due to temporal autocorrelation between neighboring EEG windows [13]. The within-domain baseline estimates performance when training and test data share the same dataset and condition composition but originate from different recording units. The remaining protocols examine whether discriminative performance is retained when training and testing are separated by participant, recording session, task composition, or dataset. These protocols were applied only where the structure of each release supported their interpretation: repeated recording days in the mental attention-state dataset enabled cross-session validation, whereas multiple tasks in SAM-40 enabled cross-task validation. Model fitting was confined to the training data, and the resulting decoder was evaluated on held-out data without target-specific adaptation. Any class imbalance was handled by oversampling the minority class within the training data only, while the test data retained their observed class distribution.

Baseline (within-dataset, within-subject evaluation). Reference performance was estimated using five-fold stratified group cross-validation, separately for the mental attention-state dataset, the complete SAM-40 composition, and each single-task composition of SAM-40. The grouping variable was the physical recording unit: one selected session in the mental attention-state dataset and one run-level recording unit in SAM-40, comprising all task recordings included in the analysis for that run. Training and test folds thus shared the dataset and condition composition and could contain data from the same participants, but never from the same recording unit. These estimates serve as within-domain comparators for the transfer protocols. No separate baseline was defined for the two-task composite compositions.

Cross-subject validation. In each leave-one-subject-out fold, the decoder was trained on all selected data from the remaining participants and tested on all selected data from the held-out participant. The resulting performance quantifies zero-shot generalization to an unseen participant.

Cross-session validation. For each participant in the mental attention-state dataset, the decoder was trained on the selected sessions other than one and tested on the held-out session. Participant identity, task protocol, and acquisition system were therefore held constant while the recording day changed. Although individual recordings in SAM-40 are distributed as separate files, the source article describes the three repetitions as trials and does not identify them as distinct recording sessions. SAM-40 was therefore not included in this analysis.

Cross-task validation. In SAM-40, each task composition represented the same binary contrast of relaxation versus one or two high-demand tasks, rather than classification of task identity. Cross-task transfer therefore asked whether a decoder fitted to this contrast under one task composition retained discriminative performance under another. We evaluated both directions between the three single-task compositions and between single-task and complementary two-task compositions. Single-task targets were interpreted relative to their corresponding within-domain baselines.

The cross-task folds were also subject-disjoint. For each fold, the source composition from all but one participant formed the training set, whereas the target composition from the held-out participant formed the test set. The protocol thus evaluates the combined shift to an unseen participant and a different task composition, rather than within-participant transfer between tasks. This distinction prevents participant-specific EEG patterns from contributing to apparent cross-task generalization [24, 25].

Cross-dataset validation. To evaluate transfer between datasets, models were trained on all available data from a source composition and tested on a composition from the other dataset, in both transfer directions and without target-specific adaptation. SAM-40 was considered as a complete composition and through its single-task and two-task compositions. Unlike the preceding protocols, cross-dataset validation changes several factors simultaneously, including the participant population, experimental paradigm, recording setup, and condition definitions. It therefore measures transfer under their compound distribution shift rather than attributing a performance change to any single factor [15]. The common sampling rate, harmonized binary labels, and shared 12-channel montage establish a compatible input and prediction space, but do not remove the substantive differences between the two datasets.

Data preparation

A common, deliberately parsimonious preparation pipeline, implemented in MNE-Python, was applied to both datasets and across all validation protocols. Its filtering, ICA, channel alignment and windowing parameters were fixed in advance and were not tuned separately for a dataset, task composition, validation protocol or decoder. The purpose was to place recordings in a common input space while limiting preprocessing itself as an additional source of variation. Differences between validation results are therefore interpreted against a fixed preparation procedure.

Recording and label selection. Recordings were selected before signal processing. Following the source study, the first two recording sessions for each participant in the mental attention-state dataset were intended for habituation to the simulator and were not included in the analyses [12]. The *Drowsy* condition was excluded from the binary classification analysis because participants were permitted to close their eyes

and enter a drowsy state. This condition differs from the *Focused* and *Unfocused* conditions not only in protocol-defined task demand, but also in arousal, visual input and ocular activity. Including it in the low-attention class would therefore confound the intended contrast, so no drowsy epochs were generated or presented to a classifier. In contrast, the complete collection of SAM-40 recordings was retained for analysis. SAM-40 provides the task recordings of each trial as separate files. Within each scenario, the task recordings selected from one trial were grouped into a single recording unit, whereas task recordings not selected by the scenario were not carried forward to later preparation steps. Accordingly, the binary mapping in Table 1 was applied to the retained recordings from the two datasets.

Filtering and ICA cleaning. Processing was performed for each recording unit, defined by the combination of participant, session and run identifiers. To prevent preprocessing from becoming an additional source of variation, we used a deliberately restrained, conventional procedure: standard filtering followed by ICA-based artifact cleaning. Keeping preparation deliberately simple allows differences in performance to be interpreted primarily in relation to the recordings and validation protocols, rather than to a complex preparation pipeline. Within each recording unit, EEG channels were notch-filtered at 50 Hz, band-pass filtered from 0.5 to 45 Hz and rereferenced to the average reference. Artifact removal used FastICA fitted to the concatenated recordings of each recording unit, retaining components that explained 99% of the variance. The decomposition was fitted to a 1–45 Hz copy of the prepared signal and then applied to each recording separately, while the analysis signal retained the 0.5–45 Hz band-pass. Ocular components were identified with F3, F4, F7 and F8 as EOG-proxy channels at a z-score threshold of 2.4 and removed. The ICA random state was fixed by the experiment configuration. When an excluded condition occurred as a block within an otherwise selected continuous recording, it remained available for ICA fitting but was omitted from window generation and the analysis sets. In the mental attention-state dataset, this applied to the *Drowsy* block.

Channel alignment and window segmentation. After signal cleaning, each recording was restricted to the 12 channels shared by the two datasets: F3, F4, F7, F8, FC5, FC6, O1, O2, P7, P8, T7 and T8. Each retained labelled block was segmented into 5-s windows with 0.5-s overlap. Every retained window inherited the label of its source block and metadata identifying its dataset, participant, session, task, run, recording unit and temporal onset. The recording-unit identifier was used by the validation protocols to ensure that all overlapping windows from a unit remained on the same side of a split.

Models

We evaluated five classifiers rather than relying on a single decoder to test whether the conclusions are robust to a change in modelling assumptions. This makes the decoder an explicit factor of the design: a

Table 2. Analytical dataset compositions. All Dataset B compositions contain the same 40 participants; they differ only in the task recordings retained with the resting-state recordings. Window counts refer to the prepared, pre-oversampling data.

Composition	Retained conditions	Participants	Recording units	Windows, n (low / high)	Class distribution, % (low / high)
Full Dataset A	Focused, Unfocused	5	24	3,192 / 3,192	50.0 / 50.0
Full Dataset B	Rest, Stroop, Arithmetic, Mirror	40	120	600 / 1,800	25.0 / 75.0
Dataset B: Stroop	Rest, Stroop	40	120	600 / 600	50.0 / 50.0
Dataset B: Arithmetic	Rest, Arithmetic	40	120	600 / 600	50.0 / 50.0
Dataset B: Mirror	Rest, Mirror	40	120	600 / 600	50.0 / 50.0
Dataset B: Stroop + Arithmetic	Rest, Stroop, Arithmetic	40	120	600 / 1,200	33.3 / 66.7
Dataset B: Stroop + Mirror	Rest, Stroop, Mirror	40	120	600 / 1,200	33.3 / 66.7
Dataset B: Arithmetic + Mirror	Rest, Arithmetic, Mirror	40	120	600 / 1,200	33.3 / 66.7

difference between validation protocols that is reproduced across models is more credibly attributed to the data and evaluation protocol than to the idiosyncrasies of one learner. The selected models span three major families used in EEG decoding: linear and tree-based traditional machine-learning models, and deep-learning architectures. We place particular emphasis on deep learning because these models learn task-relevant representations directly from the channel-by-time EEG windows, reducing reliance on a manually specified feature set. Hyperparameters were fixed *a priori* for each model and held constant across validation protocols and experimental runs; they were not tuned to an individual protocol or run and are recorded in full in the configuration stored with every run. All models were trained to predict whether a window belonged to a high-demand or a low-demand condition in the sense of Table 1, and not to predict a level of attention.

Traditional machine learning. Logistic regression provides a transparent linear baseline and represents the simplest decision rule supported by the fixed feature representation. XGBoost [26] represents tree-based ensemble learning, allowing nonlinear relations and feature interactions to be modelled within the same representation. It was also the decoder used in the originally submitted version of this study, preserving a direct point of comparison with those results. Both models received a fixed, label-free feature representation. The features were selected to provide complementary and interpretable summaries of each EEG window: band power describes its spectral distribution across conventional frequency ranges, entropy features characterize signal irregularity and complexity, and descriptive statistics characterize its amplitude distribution. The same prespecified feature set was used in every scenario rather than being selected or tuned separately for a dataset or validation protocol. It comprised log band power in delta (0.5–4 Hz), theta (4–8 Hz), alpha (8–13 Hz), beta (13–30 Hz) and gamma (30–45 Hz), together with spectral, approximate, sample and singular-value-decomposition entropy, mean, standard deviation, variance, skewness and kurtosis, computed for each channel. This produced 168 features per window (12 channels $\times$ 14 features per channel). Feature extraction contained no trainable transformation. Transformations that learn from data were fitted within each training fold only: logistic-regression features were standardized on the training data, whereas XGBoost received unscaled features.

Deep learning. The remaining three classifiers operate directly on the prepared channel-by-time EEG

windows and learn their representations during training. *ShallowFBCSPNet* [27] is a shallow convolutional neural network designed to emulate the main operations of the filter-bank common spatial pattern (FBCSP) pipeline within a trainable model. It first applies temporal convolution and spatial filtering across channels, then uses a squaring nonlinearity, mean pooling and a logarithmic activation to form band-power-like features. Thus, its inductive bias reflects a conventional EEG decoding approach, while its filters are optimized jointly with the classifier.

EEGNet [28] is a compact convolutional neural network designed specifically for EEG. Its temporal convolutions learn filters over time, depthwise convolutions learn spatial filters across electrodes for each temporal feature map, and separable convolutions combine these learned features efficiently. This factorized design substantially limits the number of trainable parameters while retaining separate temporal and spatial feature learning, making EEGNet the smallest network in the present comparison.

EEG Conformer [29] combines a convolutional front end with a Transformer encoder. The convolutional module forms local temporal and spatial features and converts them into a sequence of embeddings; self-attention in the Transformer then models global dependencies between these embeddings. It therefore complements the local receptive fields of convolution with a mechanism for relating EEG patterns over longer time spans. The only fixed input transformation for the deep-learning models was conversion from volts to microvolts. The networks were trained from scratch. No pretrained weights were used.

[TODO — Which decoder carries the Results narrative?]

Results

Balanced accuracy was the primary measure because it weights the two protocol-defined classes equally despite the unequal class frequencies in several compositions. For each scenario, subject-level balanced accuracies were averaged with equal weight per target participant; the intervals in Figs. 1 and 2 are 95% percentile bootstrap intervals over participants. Table 3 and Fig. 3 report the corresponding estimates pooled over test windows, with macro F1, MCC, ROC-AUC and per-class recall also given in the table. The complete results for all five decoders are provided in S1 Table–S5 Table; subject-level plots for the three neural networks are provided in S1 Fig–S3 Fig. Unless specified otherwise, the ranges below refer to the equally weighted subject-level estimates.

Table 3. Performance metrics across experimental scenarios for logistic regression.

Scenario	Validation method	Balanced Accuracy	Macro F1	MCC	ROC AUC	Per-class Recall (Low/High)
Baseline (within)	Stratified K-fold(Full A)	0.64	0.64	0.28	0.71	0.62/0.66
	Stratified K-fold(Full B)	0.68	0.65	0.33	0.73	0.65/0.71
	Stratified K-fold(B: Stroop)	0.69	0.69	0.39	0.74	0.67/0.71
	Stratified K-fold(B: Arithmetic)	0.66	0.66	0.33	0.72	0.66/0.67
	Stratified K-fold(B: Mirror)	0.69	0.69	0.37	0.73	0.66/0.72
Cross-subject	Leave-one-subject-out(Full A)	0.63	0.63	0.27	0.69	0.63/0.64
	Leave-one-subject-out(Full B)	0.66	0.63	0.29	0.71	0.63/0.69
Cross-session	Leave-one-session-out(Full A)	0.63	0.63	0.26	0.67	0.55/0.70
Cross-task (Dataset B)	Stroop → Arithmetic	0.59	0.59	0.19	0.63	0.65/0.54
	Stroop → Mirror	0.66	0.66	0.33	0.70	0.67/0.66
	Arithmetic → Stroop	0.56	0.56	0.12	0.60	0.67/0.46
	Arithmetic → Mirror	0.61	0.58	0.20	0.65	0.65/0.56
	Mirror → Stroop	0.62	0.62	0.25	0.68	0.63/0.62
	Mirror → Arithmetic	0.60	0.60	0.19	0.63	0.66/0.54
	Arithmetic+Mirror → Stroop	0.61	0.58	0.20	0.65	0.65/0.56
	Stroop+Mirror → Arithmetic	0.66	0.66	0.33	0.72	0.59/0.74
	Stroop+Arithmetic → Mirror	0.62	0.62	0.25	0.67	0.62/0.63
	Stroop → Arithmetic+Mirror	0.58	0.57	0.16	0.60	0.65/0.50
	Arithmetic → Stroop+Mirror	0.60	0.58	0.20	0.63	0.63/0.58
	Mirror → Stroop+Arithmetic	0.63	0.63	0.25	0.66	0.64/0.62
	Full A → Full B	0.52	0.39	0.04	0.54	0.78/0.26
	Full B → Full A	0.55	0.53	0.12	0.58	0.35/0.76
	Full A → Stroop	0.49	0.46	-0.03	0.50	0.71/0.27
	Stroop → Full A	0.54	0.54	0.08	0.56	0.45/0.63
	Full A → Arithmetic	0.53	0.52	0.07	0.56	0.68/0.39
	Arithmetic → Full A	0.53	0.52	0.05	0.54	0.44/0.61
	Full A → Mirror	0.53	0.51	0.07	0.54	0.72/0.34
	Mirror → Full A	0.53	0.49	0.06	0.55	0.26/0.80
Cross-dataset	Full A → Stroop+Arithmetic	0.51	0.44	0.02	0.53	0.72/0.30
	Stroop+Arithmetic → Full A	0.52	0.51	0.05	0.54	0.36/0.68
	Full A → Stroop+Mirror	0.51	0.43	0.02	0.52	0.75/0.27
	Stroop+Mirror → Full A	0.53	0.50	0.06	0.55	0.28/0.78
	Full A → Arithmetic+Mirror	0.54	0.46	0.08	0.56	0.77/0.30
	Arithmetic+Mirror → Full A	0.55	0.51	0.11	0.59	0.28/0.81

Note. Per-class recall is reported as Low/High. MCC, Matthews correlation coefficient; ROC-AUC, area under the receiver operating characteristic curve. Dashes indicate values to be filled.

Within-domain discrimination was modest and depended on task composition

To establish the target-specific reference against which transfer could be evaluated, each decoder was first trained and tested within the same dataset and condition composition. Across the five decoders and five baseline compositions, balanced accuracy ranged from 0.54 to 0.71 (Fig. 3). Dataset B baselines differed by task: the ranges across decoders were 0.58–0.71 for Stroop, 0.54–0.66 for Arithmetic and 0.60–0.68 for Mirror.

Thus, even before introducing a transfer shift, the separability of the high-demand versus low-demand protocol contrast was neither high nor constant across task compositions.

Within-dataset generalization depended on both the held-out factor and decoder

To determine how much discrimination remained when recordings differed within a dataset, we next held out participants or sessions. The cross-subject estimates ranged from 0.55 to 0.64 across decoders and datasets (Fig. 2) and were below their matched within-domain baseline in nine of ten decoder–dataset comparisons. In contrast, the cross-session estimates for Dataset A occupied a narrow range (0.57–0.61). These distinct patterns show that participant and session holdouts were not interchangeable sources of distribution shift.

Cross-task transfer retained only part of target-specific discrimination

To test transfer across task compositions without allowing participant identity to bridge source and target, cross-task evaluation combined a task change with a held-out participant. Balanced accuracy ranged from 0.54 to 0.65 across the 60 decoder–direction combinations. For the 45 combinations whose single-task target had a matched baseline, 42 were lower than that baseline; the mean change was -0.04 points and individual changes ranged from -0.12 to $+0.04$. The highest mean cross-task performance was obtained with ShallowFBCSPNet (0.61 across directions), but no direction or decoder retained the target-specific baseline consistently. These results indicate partial, direction-dependent transfer of the operational binary contrast across tasks.

Cross-dataset transfer produced the largest and most consistent loss

Finally, transfer between datasets was evaluated after aligning sampling rate, labels and the 12 shared channels. Cross-dataset balanced accuracy ranged from 0.47 to 0.58, with an average of 0.52 across all decoders and directions. Every one of the 55 directed comparisons for which a matched target-specific baseline was available was lower than that baseline. The loss ranged from 0.01 to 0.22 balanced-accuracy points and averaged 0.09. Transfer was also asymmetric in the matched full-composition pair: every decoder scored higher from full Dataset B to full Dataset A (0.51–0.56) than in the reverse direction (0.49–0.51). Near-chance balanced accuracy often coincided with strongly unequal class recall. For example, EEGNet trained on full Dataset A and tested on full Dataset B recalled 0.93 of low-demand but only 0.08 of high-demand windows (S3 Table). Hence, the cross-dataset loss reflected both reduced overall discrimination and, for some models, a pronounced shift toward one class.

No decoder removed the protocol effect

Across all 35 scenarios, the best point estimate changed with the scenario and the difference between the highest and lowest decoder estimate ranged from 0.01 to 0.13 (Fig. 3). ShallowFBCSPNet provided the highest estimate in 16 scenarios, but its mean cross-dataset balanced accuracy remained 0.52. Logistic regression, XGBoost, EEGNet and EEGConformer likewise averaged 0.51–0.53 under cross-dataset transfer. The common deterioration across linear, tree-based, convolutional and attention-based decoders therefore formed the principal take-home pattern of the benchmark.

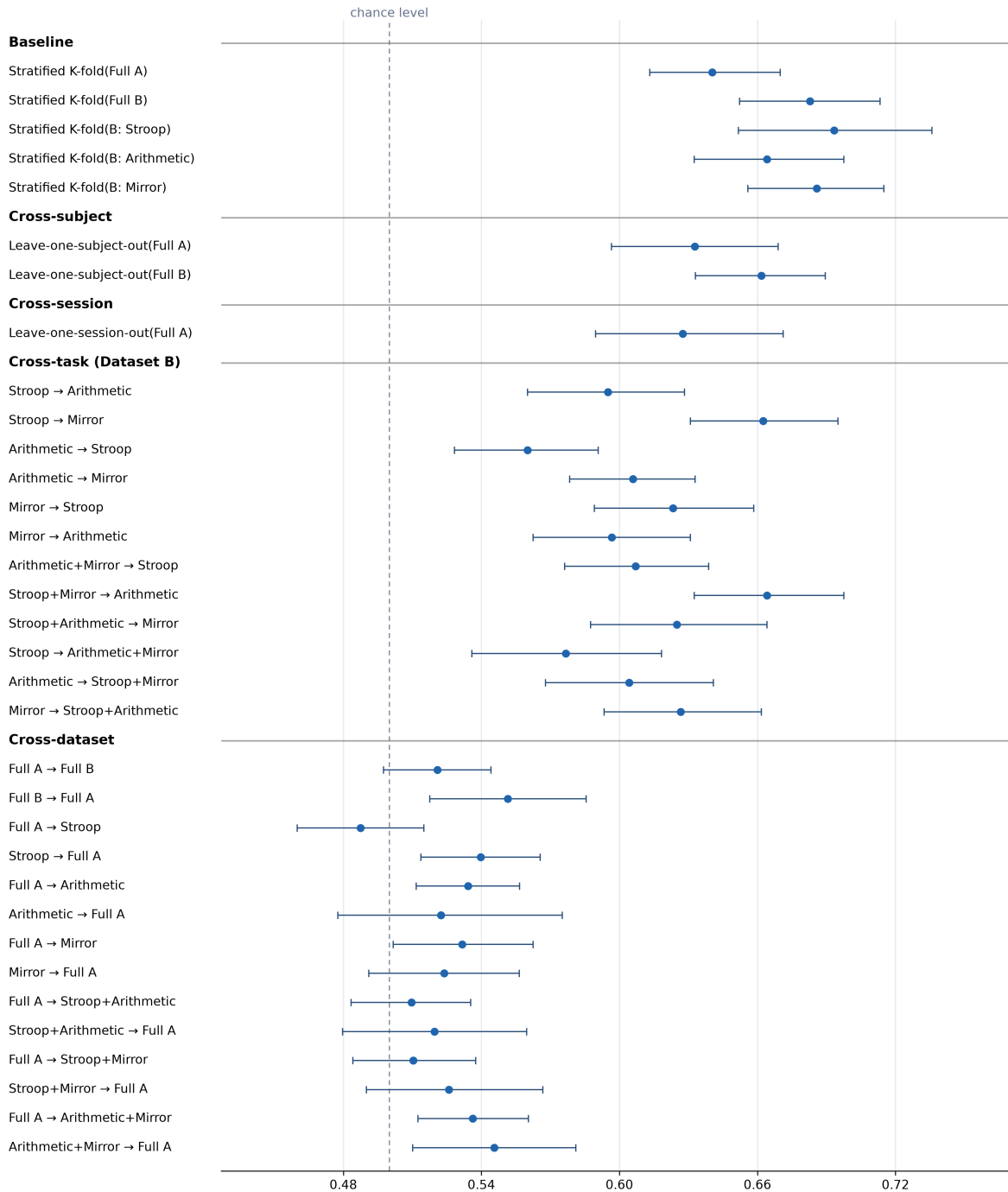


Fig 1. Subject-level balanced accuracy across validation protocols for spectral logistic regression. Points are means of target-participant balanced accuracies, with each participant weighted equally; horizontal bars are 95% percentile bootstrap intervals over participants (10,000 resamples). The dashed vertical line marks chance-level balanced accuracy (0.50). Results are grouped by baseline, cross-subject, cross-session, cross-task and cross-dataset protocols.

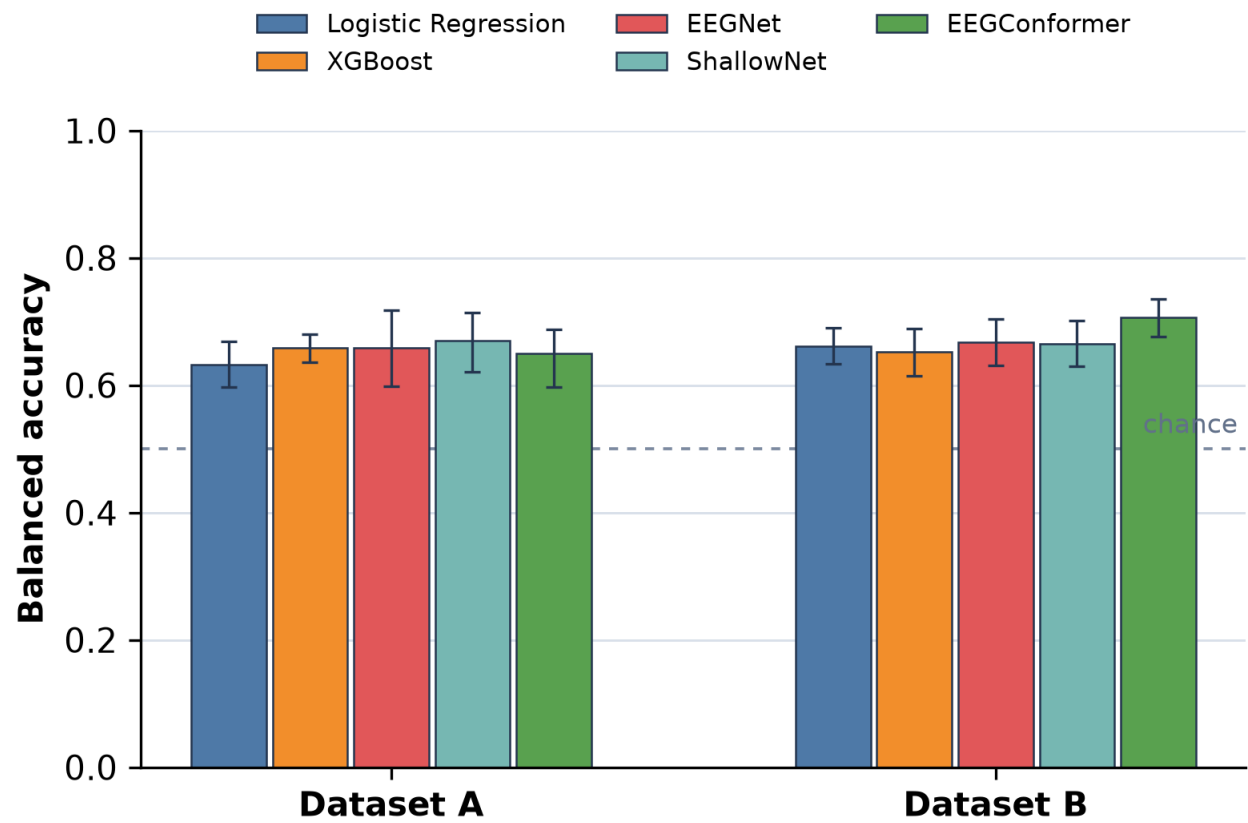


Fig 2. Cross-subject balanced accuracy by decoder and dataset. Points are means of held-out-participant balanced accuracies and error bars are 95% percentile bootstrap intervals over participants (10,000 resamples). The dashed line marks chance-level balanced accuracy (0.50).

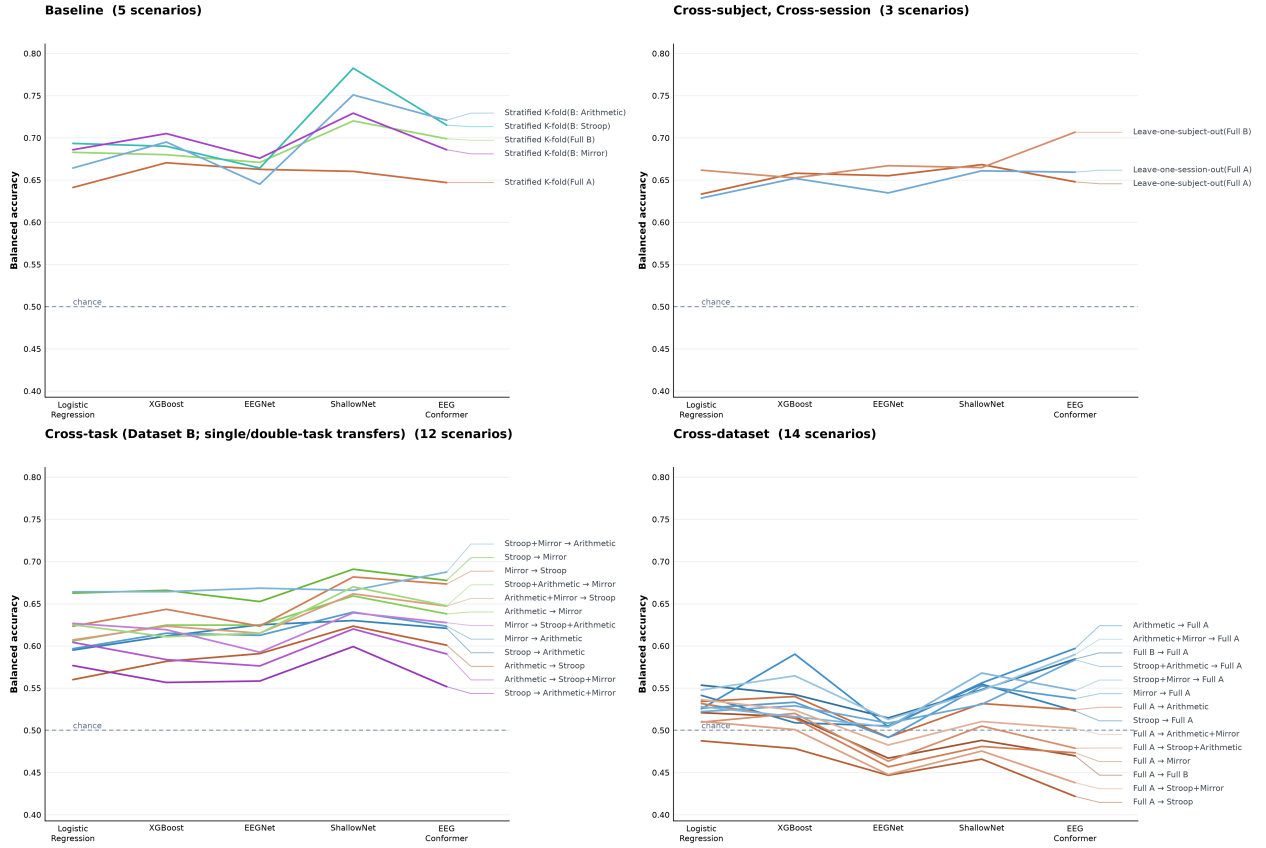


Fig 3. Pooled balanced accuracy across five decoders for every validation scenario. Each line follows one scenario across logistic regression, XGBoost, EEGNet, ShallowFBCSPNet and EEGConformer; estimates pool all test windows in the corresponding scenario. Panels group baseline, within-dataset held-out, cross-task and cross-dataset scenarios. The dashed horizontal line marks chance-level balanced accuracy (0.50).

Discussion

This study shows that zero-shot transferability of the protocol-defined high-demand versus low-demand EEG contrast was governed more consistently by the source–target shift than by decoder family. Discrimination was modest within a target composition, was partly retained across participants and tasks, and approached chance under many cross-dataset directions. Most importantly, all 55 cross-dataset comparisons with a target-specific reference performed worse than that reference. The benchmark therefore supplies the missing pre-adaptation denominator needed to determine whether a future transfer method produces a genuine gain or negative transfer [11].

The within-dataset protocols also demonstrate why “generalization” should not be treated as a single property. Cross-subject performance was below the matched baseline in nine of ten decoder–dataset comparisons, consistent with the well-established inter-participant variability of EEG. Cross-session

performance in Dataset A remained close to its baseline. These results caution against assigning a model a general “robustness” label from a single validation protocol.

Cross-task transfer preserved more discrimination than cross-dataset transfer, but it usually did not recover the corresponding target baseline. This partial retention is compatible with prior evidence that a cross-participant, cross-task vigilance-decrement model can perform above chance [24]. At the same time, the reduction relative to within-task baselines agrees with evidence that ongoing spectral EEG features are sensitive to task type and can be less cross-task-consistent than task-irrelevant event-related potentials [25]. Direct numerical comparison across these studies is inappropriate because the constructs, labels, tasks and validation units differ. The present results add a complementary observation: when participant identity is excluded from both sides of the transfer and no target adaptation is performed, task-to-task retention remains direction- and decoder-dependent. Reported gains from cross-task domain adaptation [9] should therefore be read against an equally strict no-adaptation reference.

The cross-dataset results place a sharper boundary on zero-shot use. Matching sampling rate and channels created a common input space, but it did not align participant populations, acquisition details, paradigms, sensory conditions, task demands or the meaning of the protocol-assigned classes. The uniformly higher performance from full Dataset B to full Dataset A than in the reverse direction further shows that compatibility was asymmetric, so source suitability cannot be inferred from nominal label overlap alone. Prior cross-dataset EEG studies have likewise treated technical and distributional mismatch as an adaptation problem: domain adaptation improved cross-dataset emotion recognition over an unadapted baseline [15], and signal alignment improved cross-dataset P300 transfer [10]. Our results do not test those methods, but provide the unadapted reference required to quantify comparable improvements for the present operational contrast.

Decoder comparisons reinforce this interpretation without establishing a universal model ranking. ShallowFBCSPNet produced the largest point estimate in more scenarios than the other decoders, yet it deteriorated under cross-dataset transfer to the same narrow range as logistic regression, XGBoost, EEGNet and EEGConformer. Because all models used fixed settings rather than scenario-specific optimization, these patterns show model–protocol interaction, not the inherent superiority of an architecture. The recurrence of the cross-dataset loss across five materially different decoders makes it less likely that the main pattern is peculiar to one classifier, while leaving open whether adaptation, pretraining or other architectures could reduce it.

The interpretation of these findings is necessarily limited by the labels. In both datasets, the class is assigned by the experimental protocol: it contrasts a condition with higher protocol-defined task demand

against rest or an unfocused condition. It is not an independent measurement of an individual’s level of attention. Accordingly, the results show limited transferability of this operational contrast between the available datasets and paradigms. They neither demonstrate the absence of invariant neural markers of attention nor establish an upper bound on what a different decoder could achieve. Excluding the drowsy condition avoids one important difference in arousal, eye closure and ocular activity, but it does not eliminate other differences in sensory input, task demands, stress, working-memory demands or time-on-task between the retained conditions.

Several further limitations delimit the benchmark. Only two heterogeneous open datasets were available for the cross-dataset comparison, and Dataset A contained only five participants; the participant bootstrap intervals for that dataset therefore describe limited empirical support and should not be read as precise population intervals. Harmonization required restriction to 12 shared channels, improving input compatibility while reducing spatial information and limiting the stability of ICA-based artifact removal. Each decoder used one prespecified hyperparameter configuration and random seed, so small differences among decoders may reflect optimization or training variability and do not support formal superiority claims. The analysis was descriptive and did not apply multiplicity-adjusted hypothesis tests across the many directed pairs. Finally, the study did not test domain adaptation, target-specific calibration or personalization and did not identify which signal features drove individual decisions. It therefore cannot attribute transfer failure to a particular neural, ocular, muscular or other physiological source.

Future work should evaluate adaptation gains against the direction-specific baselines reported here, repeat stochastic models across seeds, and diagnose whether errors arise from acquisition shift, condition composition or class-conditional mismatch. More decisive cross-paradigm tests will require multi-site datasets that preserve leakage-resistant, subject-disjoint evaluation while measuring the target state independently of the task used to elicit it. Concurrent behavioural performance, validated self-report and external physiological measurements would make it possible to distinguish a shared cognitive state from a shared protocol label.

Conclusion

Across five decoders and five validation protocols, zero-shot EEG discriminability depended on how source and target recordings were obtained. Cross-task transfer retained part of the target-specific discrimination, but cross-dataset transfer yielded the largest loss: all 55 matched comparisons fell below their target baseline by 0.01–0.22 balanced-accuracy points. No decoder family removed this loss, and transfer direction mattered.

These results establish a reproducible pre-adaptation benchmark for selecting source data and for
distinguishing useful adaptation from negative transfer. They apply to the available protocol-defined
high-demand versus low-demand labels, not to attention as a unitary measured state; stronger claims will
require cross-paradigm datasets with independent behavioural, self-report or physiological validation of the
target construct.

[TODO — Add acknowledgments if applicable.]

References

1. Rashid M, Sulaiman N, Abdul Majeed APP, Musa RM, Ab Nasir AF, Bari BS, et al. Current status, challenges, and possible solutions of EEG-based brain–computer interface: A comprehensive review. *Frontiers in Neurorobotics*. 2020;14:25. doi:10.3389/fnbot.2020.00025.
2. Wu D, Xu Y, Lu BL. Transfer learning for EEG-based brain–computer interfaces: A review of progress made since 2016. *IEEE Transactions on Cognitive and Developmental Systems*. 2022;14(1):4-19. doi:10.1109/TCDS.2020.3007453.
3. Sun Q, Zhou Y, Gong P, Zhang D. Attention detection using EEG signals and machine learning: A review. *Machine Intelligence Research*. 2025;22(2):219-38. doi:10.1007/s11633-024-1492-6.
4. Li H, Chen Y, Wang Y, Ni W, Zhang H. Foundation models for cross-domain EEG analysis application: A survey. *arXiv preprint arXiv:250815716*. 2025. arXiv:2508.15716. doi:10.48550/arXiv.2508.15716.
5. Zhang K, Xu G, Zheng X, Li H, Zhang S, Yu Y, et al. Application of transfer learning in EEG decoding based on brain–computer interfaces: A review. *Sensors*. 2020;20(21):6321. doi:10.3390/s20216321.
6. Hommel B, Chapman CS, Cisek P, Neyedli HF, Song JH, Welsh TN. No one knows what attention is. *Attention, Perception, & Psychophysics*. 2019;81(7):2288-303. doi:10.3758/s13414-019-01846-w.
7. Knudsen EI. Fundamental Components of Attention. *Annual Review of Neuroscience*. 2007;30:57-78. doi:10.1146/annurev.neuro.30.051606.094256.
8. Petersen SE, Posner MI. The Attention System of the Human Brain: 20 Years After. *Annual Review of Neuroscience*. 2012;35:73-89. doi:10.1146/annurev-neuro-062111-150525.

9. Zhou Y, Xu Z, Niu Y, Wang P, Wen X, Wu X, et al. Cross-Task Cognitive Workload Recognition Based on EEG and Domain Adaptation. *IEEE Transactions on Neural Systems and Rehabilitation Engineering*. 2022;30:50-60. doi:10.1109/TNSRE.2022.3140456.
10. Song M, Gwon D, Jun SC, Ahn M. Signal alignment for cross-datasets in P300 brain-computer interfaces. *Journal of Neural Engineering*. 2024;21(3):036007. doi:10.1088/1741-2552/ad430d.
11. Zhang W, Deng L, Zhang L, Wu D. A Survey on Negative Transfer. *IEEE/CAA Journal of Automatica Sinica*. 2023;10(2):305-29. doi:10.1109/JAS.2022.106004.
12. Acı Çİ, Kaya M, Mishchenko Y. Distinguishing mental attention states of humans via an EEG-based passive BCI using machine learning methods. *Expert Systems with Applications*. 2019;134:153-66. doi:10.1016/j.eswa.2019.05.057.
13. Chen X, Bao X, Jitian K, Li R, Zhu L, Kong W. Hybrid EEG Feature Learning Method for Cross-Session Human Mental Attention State Classification. *Brain Sciences*. 2025;15(8):805. doi:10.3390/brainsci15080805.
14. Ghosh R, Deb N, Sengupta K, Phukan A, Choudhury N, Kashyap S, et al. SAM 40: Dataset of 40 subject EEG recordings to monitor the induced-stress while performing Stroop color-word test, arithmetic task, and mirror image recognition task. *Data in Brief*. 2022;40:107772. doi:10.1016/j.dib.2021.107772.
15. Lan Z, Sourina O, Wang L, Scherer R, Müller-Putz GR. Domain adaptation techniques for EEG-based emotion recognition: A comparative study on two public datasets. *IEEE Transactions on Cognitive and Developmental Systems*. 2019;11(1):85-94. doi:10.1109/TCDS.2018.2826840.
16. MacLeod CM. Half a century of research on the Stroop effect: An integrative review. *Psychological Bulletin*. 1991;109(2):163-203. doi:10.1037/0033-2909.109.2.163.
17. Scarpina F, Tagini S. The Stroop color and word test. *Frontiers in Psychology*. 2017;8:557. doi:10.3389/fpsyg.2017.00557.
18. Esterman M, Noonan SK, Rosenberg M, DeGutis J. In the zone or zoning out? Tracking behavioral and neural fluctuations during sustained attention. *Cerebral Cortex*. 2013;23(11):2712-23. doi:10.1093/cercor/bhs261.

19. Grabner RH, Ansari D, Reishofer G, Stern E, Ebner F, Neuper C. Individual differences in mathematical competence predict parietal brain activation during mental calculation. *NeuroImage*. 2007;38(2):346-56. doi:10.1016/j.neuroimage.2007.07.041.
20. Kitaura Y, Nishida K, Yoshimura M, Mii H, Katsura K, Ueda S, et al. Functional localization and effective connectivity of cortical theta and alpha oscillatory activity during an attention task. *Clinical Neurophysiology Practice*. 2017;2:193-200. doi:10.1016/j.cnp.2017.09.002.
21. Shepard RN, Metzler J. Mental rotation of three-dimensional objects. *Science*. 1971;171(3972):701-3. doi:10.1126/science.171.3972.701.
22. Hyun JS, Luck SJ. Visual working memory as the substrate for mental rotation. *Psychonomic Bulletin & Review*. 2007;14(1):154-8. doi:10.3758/BF03194043.
23. Tomasino B, Gremese M. Effects of stimulus type and strategy on mental rotation network: An activation likelihood estimation meta-analysis. *Frontiers in Human Neuroscience*. 2016;9:693. doi:10.3389/fnhum.2015.00693.
24. Kamrud A, Borghetti B, Schubert Kabban C, Miller M. Generalized deep learning EEG models for cross-participant and cross-task detection of the vigilance decrement in sustained attention tasks. *Sensors*. 2021;21(16):5617. doi:10.3390/s21165617.
25. Ke Y, Jiang T, Liu S, Cao Y, Jiao X, Jiang J, et al. Cross-task consistency of electroencephalography-based mental workload indicators: Comparisons between power spectral density and task-irrelevant auditory event-related potentials. *Frontiers in Neuroscience*. 2021;15:703139. doi:10.3389/fnins.2021.703139.
26. Chen T, Guestrin C. XGBoost: a scalable tree boosting system. In: *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*; 2016. p. 785-94. doi:10.1145/2939672.2939785.
27. Schirrneister RT, Springenberg JT, Fiederer LDJ, Glasstetter M, Eggersperger K, Tangermann M, et al. Deep learning with convolutional neural networks for EEG decoding and visualization. *Human Brain Mapping*. 2017;38(11):5391-420. doi:10.1002/hbm.23730.
28. Lawhern VJ, Solon AJ, Waytowich NR, Gordon SM, Hung CP, Lance BJ. EEGNet: a compact convolutional neural network for EEG-based brain-computer interfaces. *Journal of Neural Engineering*. 2018;15(5):056013. doi:10.1088/1741-2552/aace8c.

29. Song Y, Zheng Q, Liu B, Gao X. EEG Conformer: convolutional transformer for EEG decoding and visualization. *IEEE Transactions on Neural Systems and Rehabilitation Engineering*. 2023;31:710-9. doi:10.1109/TNSRE.2022.3230250.

Supporting information

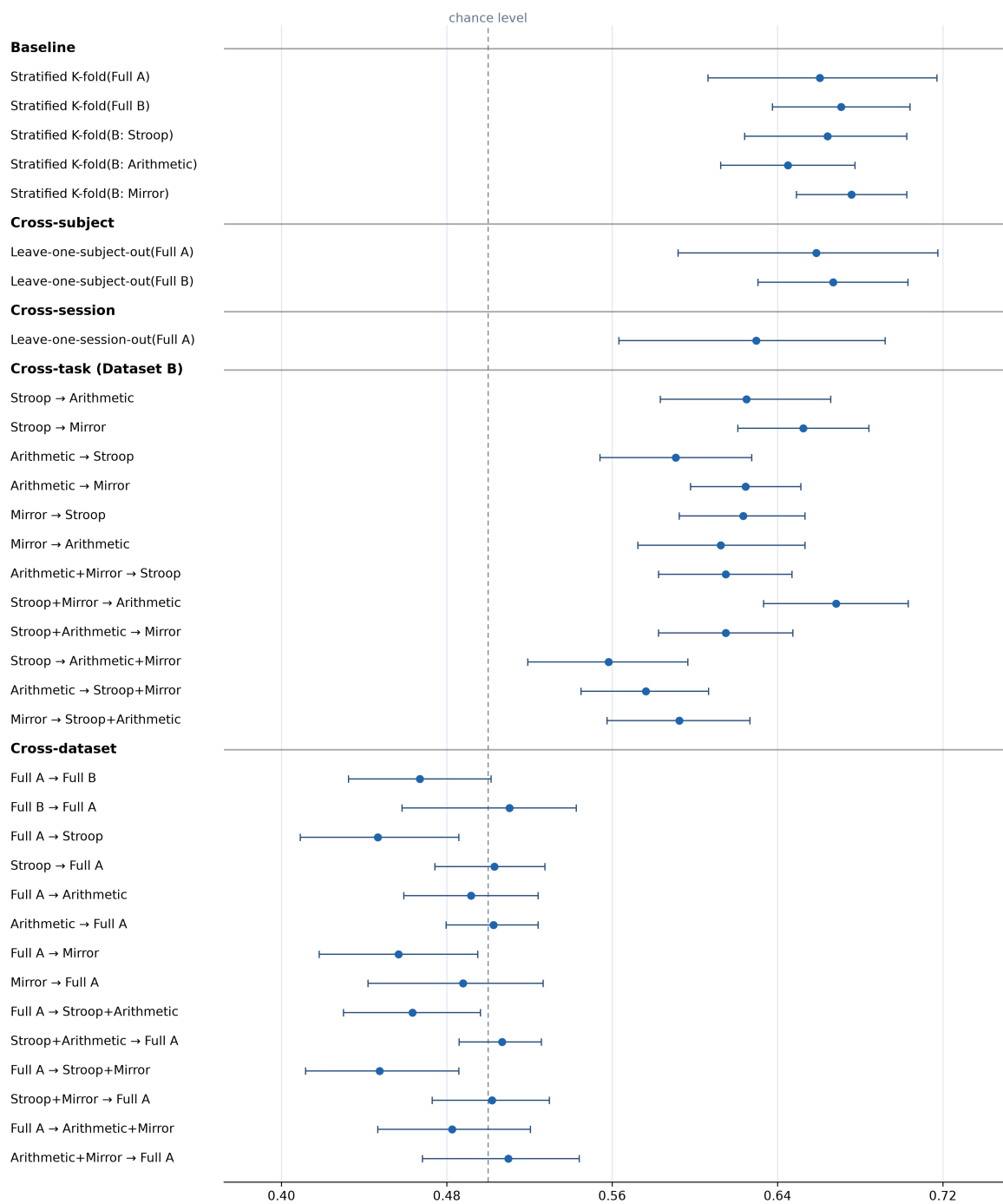
S1 Fig. EEGNet balanced accuracy across validation protocols. The per-protocol EEGNet plot uses the same point, interval and chance-line conventions as Fig. 1.

S2 Fig. ShallowFBCSPNet balanced accuracy across validation protocols. The per-protocol ShallowFBCSPNet plot uses the same point, interval and chance-line conventions as Fig. 1.

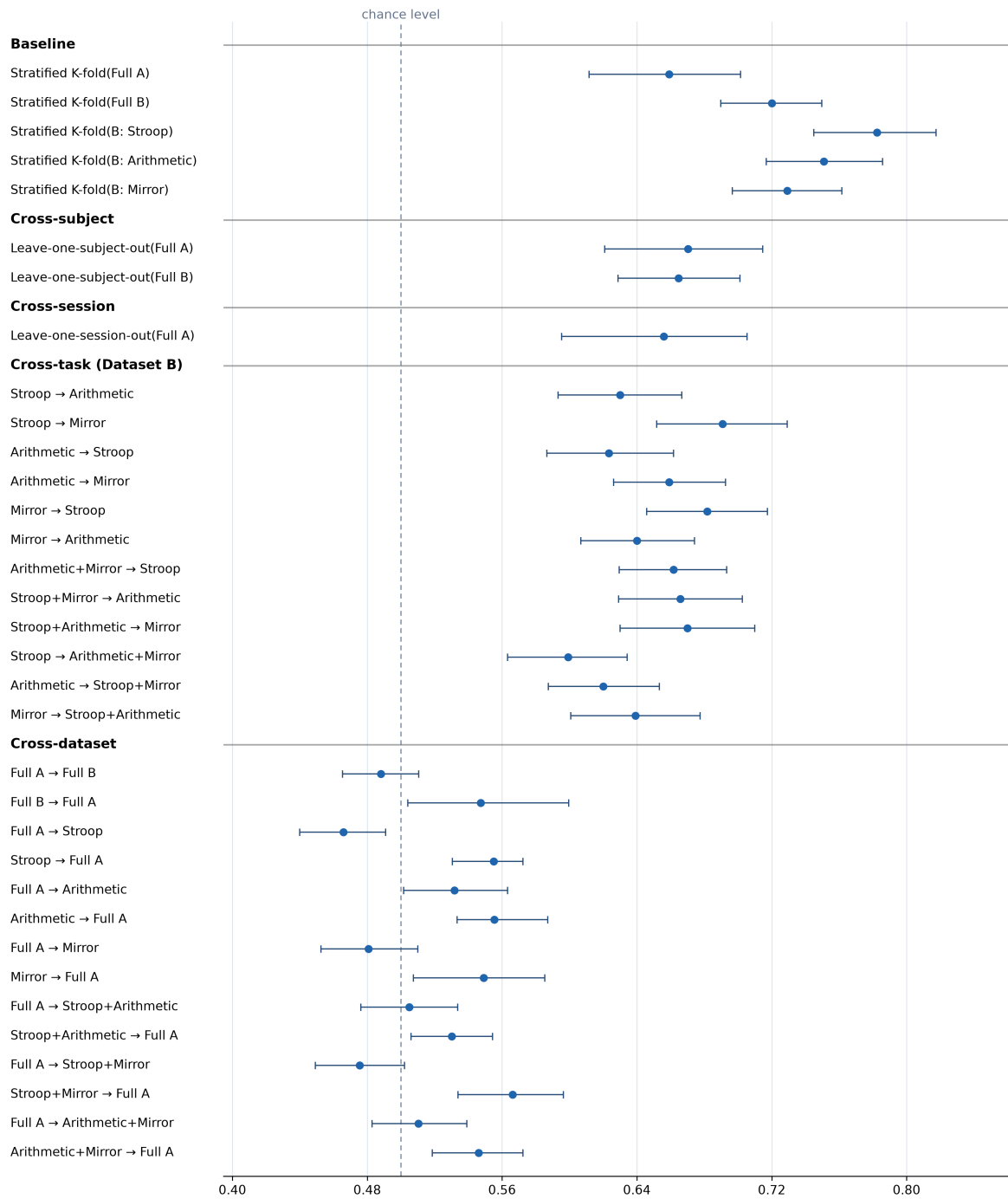
S3 Fig. EEGConformer balanced accuracy across validation protocols. The per-protocol EEGConformer plot uses the same point, interval and chance-line conventions as Fig. 1.

Supporting-information figure previews

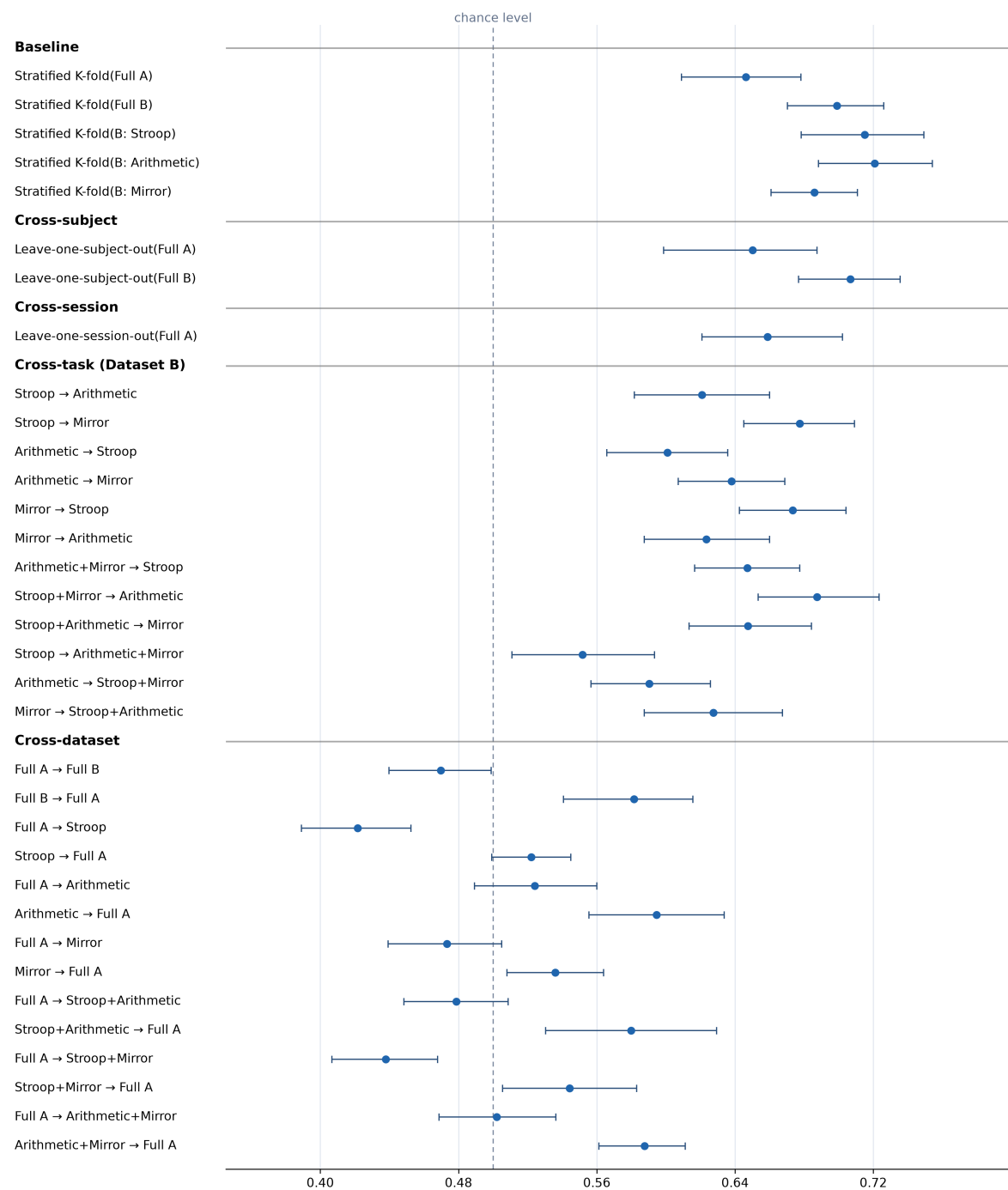
S1 Fig. EEGNet balanced accuracy across validation protocols.



S2 Fig. ShallowFBCSPNet balanced accuracy across validation protocols.



S3 Fig. EEGConformer balanced accuracy across validation protocols.



S1 Table. Logistic-regression performance metrics across all experimental scenarios.

S2 Table. XGBoost performance metrics across all experimental scenarios.

S3 Table. EEGNet performance metrics across all experimental scenarios.

S4 Table. ShallowFBCSPNet performance metrics across all experimental scenarios.

S5 Table. EEGConformer performance metrics across all experimental scenarios.

S6 Table. Preprocessing, feature extraction and model configuration. [TODO — Regenerate this table for the current model set.]

Full parameter values for every preprocessing, feature extraction and model step, as recorded in the versioned experiment configuration. Continuous-signal filtering: band-pass 0.5–45 Hz, 50 Hz notch, average reference. Artifact removal: FastICA, 0.99 explained-variance component count, fit on data band-passed 1–45 Hz, EOG-proxy channels F3/F4/F7/F8 with threshold 2.4, random state 42. Epoching: 5 s windows with 0.5 s overlap over a common 12-channel montage (F3, F4, F7, F8, FC5, FC6, O1, O2, P7, P8, T7, T8). Record selection: sessions with `session_role=habituation` or `session_quality=incomplete` excluded prior to Stage 1. Feature extraction (spectral configuration): log band power in five bands (0.5–4, 4–8, 8–13, 13–30, 30–45 Hz) plus spectral entropy per channel. Models: the five configurations named in the Models subsection, with the feature scaler fitted inside the pipeline on the training fold only. Global random seed: 42. Validation protocols: stratified group K -fold baseline grouped by physical recording, cross-subject, cross-session (session-documented dataset only), cross-dataset and leave-one-subject-out cross-task transfer. No data-driven hyperparameter tuning was performed. Hyperparameters were fixed a priori for the benchmark comparison.

Supporting-information table previews

Table 4. Performance metrics across experimental scenarios for XGBoost.

Scenario	Validation method	Balanced Accuracy	Macro F1	MCC	ROC AUC	Per-class Recall (Low/High)
Baseline (within)	Stratified K-fold(Full A)	0.67	0.67	0.34	0.73	0.64/0.70
	Stratified K-fold(Full B)	0.68	0.66	0.33	0.74	0.60/0.76
	Stratified K-fold(B: Stroop)	0.69	0.69	0.38	0.76	0.68/0.70
	Stratified K-fold(B: Arithmetic)	0.70	0.69	0.39	0.75	0.69/0.70
	Stratified K-fold(B: Mirror)	0.70	0.70	0.41	0.76	0.72/0.69
Cross-subject	Leave-one-subject-out(Full A)	0.66	0.66	0.32	0.72	0.64/0.67
	Leave-one-subject-out(Full B)	0.65	0.63	0.28	0.71	0.58/0.72
Cross-session	Leave-one-session-out(Full A)	0.65	0.65	0.31	0.70	0.53/0.77
Cross-task (Dataset B)	Stroop → Arithmetic	0.61	0.61	0.22	0.64	0.65/0.58
	Stroop → Mirror	0.67	0.67	0.33	0.72	0.66/0.67
	Arithmetic → Stroop	0.58	0.58	0.17	0.62	0.68/0.48
	Arithmetic → Mirror	0.62	0.60	0.23	0.67	0.67/0.58
	Mirror → Stroop	0.64	0.64	0.29	0.69	0.70/0.59
	Mirror → Arithmetic	0.61	0.61	0.24	0.65	0.72/0.51
	Arithmetic+Mirror → Stroop	0.62	0.59	0.23	0.67	0.70/0.55
	Stroop+Mirror → Arithmetic	0.66	0.66	0.33	0.72	0.60/0.72
	Stroop+Arithmetic → Mirror	0.61	0.61	0.22	0.66	0.63/0.59
	Stroop → Arithmetic+Mirror	0.56	0.55	0.12	0.58	0.67/0.45
	Arithmetic → Stroop+Mirror	0.58	0.56	0.16	0.62	0.63/0.53
	Mirror → Stroop+Arithmetic	0.62	0.62	0.24	0.66	0.65/0.59
	Full A → Full B	0.52	0.52	0.03	0.52	0.28/0.75
	Full B → Full A	0.54	0.53	0.09	0.55	0.37/0.71
	Full A → Stroop	0.48	0.45	-0.05	0.47	0.26/0.70
	Stroop → Full A	0.51	0.49	0.02	0.51	0.33/0.69
	Full A → Arithmetic	0.54	0.50	0.10	0.55	0.27/0.81
	Arithmetic → Full A	0.59	0.59	0.18	0.62	0.62/0.56
	Full A → Mirror	0.51	0.48	0.03	0.53	0.27/0.76
Cross-dataset	Mirror → Full A	0.53	0.53	0.07	0.55	0.61/0.46
	Full A → Stroop+Arithmetic	0.52	0.52	0.04	0.51	0.29/0.75
	Stroop+Arithmetic → Full A	0.53	0.52	0.06	0.53	0.37/0.68
	Full A → Stroop+Mirror	0.50	0.50	0.00	0.50	0.27/0.73
	Stroop+Mirror → Full A	0.52	0.51	0.03	0.52	0.38/0.66
	Full A → Arithmetic+Mirror	0.52	0.52	0.05	0.54	0.27/0.78
	Arithmetic+Mirror → Full A	0.56	0.56	0.13	0.58	0.57/0.56

Note. Per-class recall is reported as Low/High. MCC, Matthews correlation coefficient; ROC-AUC, area under the receiver operating characteristic curve. Dashes indicate values to be filled.

Table 5. Performance metrics across experimental scenarios for EEGNet.

Scenario	Validation method	Balanced Accuracy	Macro F1	MCC	ROC AUC	Per-class Recall (Low/High)
Baseline (within)	Stratified K-fold(Full A)	0.66	0.66	0.33	0.74	0.57/0.76
	Stratified K-fold(Full B)	0.67	0.66	0.32	0.73	0.57/0.77
	Stratified K-fold(B: Stroop)	0.66	0.66	0.33	0.71	0.58/0.74
	Stratified K-fold(B: Arithmetic)	0.65	0.64	0.29	0.70	0.60/0.69
	Stratified K-fold(B: Mirror)	0.68	0.67	0.36	0.73	0.60/0.76
Cross-subject	Leave-one-subject-out(Full A)	0.66	0.66	0.31	0.72	0.65/0.66
	Leave-one-subject-out(Full B)	0.67	0.64	0.30	0.71	0.60/0.74
Cross-session	Leave-one-session-out(Full A)	0.63	0.63	0.27	0.67	0.59/0.68
Cross-task (Dataset B)	Stroop → Arithmetic	0.62	0.62	0.25	0.65	0.59/0.66
	Stroop → Mirror	0.65	0.65	0.31	0.69	0.60/0.71
	Arithmetic → Stroop	0.59	0.59	0.18	0.62	0.61/0.57
	Arithmetic → Mirror	0.62	0.61	0.24	0.66	0.61/0.64
	Mirror → Stroop	0.62	0.62	0.25	0.66	0.57/0.68
	Mirror → Arithmetic	0.61	0.61	0.23	0.66	0.57/0.65
	Arithmetic+Mirror → Stroop	0.61	0.61	0.22	0.66	0.57/0.66
	Stroop+Mirror → Arithmetic	0.67	0.67	0.34	0.72	0.61/0.73
	Stroop+Arithmetic → Mirror	0.61	0.61	0.23	0.66	0.60/0.63
	Stroop → Arithmetic+Mirror	0.56	0.55	0.12	0.58	0.65/0.47
	Arithmetic → Stroop+Mirror	0.58	0.56	0.14	0.60	0.59/0.56
	Mirror → Stroop+Arithmetic	0.59	0.59	0.19	0.64	0.60/0.58
	Full A → Full B	0.47	0.39	-0.06	0.47	0.61/0.32
	Full B → Full A	0.51	0.49	0.03	0.53	0.30/0.73
	Full A → Stroop	0.45	0.43	-0.11	0.43	0.60/0.29
	Stroop → Full A	0.51	0.48	0.01	0.51	0.27/0.74
	Full A → Arithmetic	0.49	0.49	-0.02	0.51	0.58/0.40
Cross-dataset	Arithmetic → Full A	0.50	0.44	0.01	0.52	0.16/0.85
	Full A → Mirror	0.46	0.45	-0.09	0.47	0.60/0.32
	Mirror → Full A	0.49	0.49	-0.02	0.48	0.44/0.54
	Full A → Stroop+Arithmetic	0.46	0.42	-0.07	0.47	0.60/0.32
	Stroop+Arithmetic → Full A	0.51	0.40	0.03	0.58	0.08/0.94
	Full A → Stroop+Mirror	0.45	0.40	-0.11	0.45	0.60/0.29
	Stroop+Mirror → Full A	0.50	0.39	0.01	0.50	0.08/0.93
	Full A → Arithmetic+Mirror	0.48	0.44	-0.03	0.49	0.60/0.37
	Arithmetic+Mirror → Full A	0.51	0.49	0.03	0.52	0.28/0.74

Note. Per-class recall is reported as Low/High. MCC, Matthews correlation coefficient; ROC-AUC, area under the receiver operating characteristic curve. Dashes indicate values to be filled.

Table 6. Performance metrics across experimental scenarios for ShallowFBCSPNet.

Scenario	Validation method	Balanced Accuracy	Macro F1	MCC	ROC AUC	Per-class Recall (Low/High)
Baseline (within)	Stratified K-fold(Full A)	0.66	0.66	0.32	0.73	0.63/0.69
	Stratified K-fold(Full B)	0.72	0.72	0.44	0.81	0.58/0.86
	Stratified K-fold(B: Stroop)	0.78	0.78	0.57	0.86	0.77/0.79
	Stratified K-fold(B: Arithmetic)	0.75	0.75	0.50	0.83	0.76/0.74
	Stratified K-fold(B: Mirror)	0.73	0.73	0.46	0.80	0.72/0.74
Cross-subject	Leave-one-subject-out(Full A)	0.67	0.67	0.34	0.73	0.72/0.62
	Leave-one-subject-out(Full B)	0.66	0.66	0.32	0.75	0.52/0.81
Cross-session	Leave-one-session-out(Full A)	0.66	0.66	0.33	0.71	0.58/0.74
Cross-task (Dataset B)	Stroop → Arithmetic	0.63	0.63	0.26	0.67	0.63/0.63
	Stroop → Mirror	0.69	0.69	0.38	0.75	0.73/0.65
	Arithmetic → Stroop	0.62	0.62	0.25	0.68	0.72/0.53
	Arithmetic → Mirror	0.66	0.63	0.30	0.72	0.73/0.59
	Mirror → Stroop	0.68	0.68	0.36	0.74	0.69/0.68
	Mirror → Arithmetic	0.64	0.64	0.28	0.68	0.72/0.56
	Arithmetic+Mirror → Stroop	0.66	0.64	0.30	0.72	0.70/0.62
	Stroop+Mirror → Arithmetic	0.67	0.66	0.33	0.74	0.61/0.72
	Stroop+Arithmetic → Mirror	0.67	0.67	0.34	0.72	0.62/0.72
	Stroop → Arithmetic+Mirror	0.60	0.59	0.20	0.65	0.72/0.48
	Arithmetic → Stroop+Mirror	0.62	0.59	0.23	0.68	0.70/0.54
	Mirror → Stroop+Arithmetic	0.64	0.64	0.28	0.70	0.73/0.55
	Full A → Full B	0.49	0.34	-0.03	0.48	0.79/0.19
	Full B → Full A	0.55	0.51	0.11	0.56	0.28/0.81
	Full A → Stroop	0.47	0.41	-0.09	0.43	0.77/0.16
	Stroop → Full A	0.55	0.55	0.11	0.58	0.52/0.59
	Full A → Arithmetic	0.53	0.51	0.07	0.55	0.74/0.32
	Arithmetic → Full A	0.56	0.56	0.11	0.59	0.55/0.56
	Full A → Mirror	0.48	0.43	-0.05	0.48	0.78/0.19
	Mirror → Full A	0.55	0.55	0.11	0.58	0.52/0.58
Cross-dataset	Full A → Stroop+Arithmetic	0.51	0.40	0.01	0.49	0.80/0.21
	Stroop+Arithmetic → Full A	0.53	0.52	0.06	0.55	0.37/0.69
	Full A → Stroop+Mirror	0.48	0.36	-0.06	0.45	0.78/0.17
	Stroop+Mirror → Full A	0.57	0.57	0.14	0.58	0.59/0.55
	Full A → Arithmetic+Mirror	0.51	0.41	0.02	0.52	0.78/0.24
	Arithmetic+Mirror → Full A	0.55	0.54	0.10	0.58	0.64/0.45

Note. Per-class recall is reported as Low/High. MCC, Matthews correlation coefficient; ROC-AUC, area under the receiver operating characteristic curve. Dashes indicate values to be filled.

Table 7. Performance metrics across experimental scenarios for EEGConformer.

Scenario	Validation method	Balanced Accuracy	Macro F1	MCC	ROC AUC	Per-class Recall (Low/High)
Baseline (within)	Stratified K-fold(Full A)	0.65	0.65	0.30	0.72	0.69/0.60
	Stratified K-fold(Full B)	0.70	0.70	0.39	0.78	0.56/0.84
	Stratified K-fold(B: Stroop)	0.72	0.71	0.44	0.79	0.59/0.84
	Stratified K-fold(B: Arithmetic)	0.72	0.72	0.45	0.79	0.61/0.83
	Stratified K-fold(B: Mirror)	0.69	0.68	0.38	0.77	0.59/0.78
Cross-subject	Leave-one-subject-out(Full A)	0.65	0.65	0.30	0.72	0.65/0.64
	Leave-one-subject-out(Full B)	0.71	0.70	0.40	0.75	0.60/0.82
Cross-session	Leave-one-session-out(Full A)	0.66	0.66	0.32	0.70	0.58/0.74
Cross-task (Dataset B)	Stroop → Arithmetic	0.62	0.62	0.24	0.66	0.65/0.59
	Stroop → Mirror	0.68	0.68	0.36	0.73	0.62/0.73
	Arithmetic → Stroop	0.60	0.60	0.20	0.64	0.61/0.59
	Arithmetic → Mirror	0.64	0.62	0.26	0.69	0.62/0.66
	Mirror → Stroop	0.67	0.67	0.35	0.72	0.64/0.71
	Mirror → Arithmetic	0.62	0.62	0.25	0.64	0.65/0.60
	Arithmetic+Mirror → Stroop	0.65	0.63	0.28	0.68	0.65/0.65
	Stroop+Mirror → Arithmetic	0.69	0.68	0.39	0.76	0.56/0.81
	Stroop+Arithmetic → Mirror	0.65	0.65	0.30	0.69	0.61/0.69
	Stroop → Arithmetic+Mirror	0.55	0.55	0.11	0.58	0.67/0.44
	Arithmetic → Stroop+Mirror	0.59	0.57	0.17	0.62	0.62/0.56
	Mirror → Stroop+Arithmetic	0.63	0.63	0.26	0.67	0.63/0.63
	Full A → Full B	0.47	0.36	-0.06	0.46	0.70/0.24
	Full B → Full A	0.58	0.58	0.17	0.62	0.62/0.55
	Full A → Stroop	0.42	0.39	-0.18	0.40	0.65/0.20
	Stroop → Full A	0.52	0.44	0.07	0.61	0.14/0.91
	Full A → Arithmetic	0.52	0.52	0.05	0.54	0.60/0.45
	Arithmetic → Full A	0.60	0.59	0.20	0.63	0.68/0.51
	Full A → Mirror	0.47	0.45	-0.06	0.46	0.66/0.29
	Mirror → Full A	0.54	0.51	0.08	0.57	0.76/0.32
Cross-dataset	Full A → Stroop+Arithmetic	0.48	0.41	-0.04	0.46	0.68/0.27
	Stroop+Arithmetic → Full A	0.58	0.58	0.17	0.62	0.58/0.58
	Full A → Stroop+Mirror	0.44	0.37	-0.13	0.43	0.65/0.23
	Stroop+Mirror → Full A	0.55	0.52	0.11	0.59	0.31/0.78
	Full A → Arithmetic+Mirror	0.50	0.44	0.00	0.50	0.68/0.32
	Arithmetic+Mirror → Full A	0.59	0.59	0.18	0.62	0.60/0.58

Note. Per-class recall is reported as Low/High. MCC, Matthews correlation coefficient; ROC-AUC, area under the receiver operating characteristic curve. Dashes indicate values to be filled.